

nowhere dense sets), which are classically equivalent to each other. However, the combinatorial definition of κ -meagre sets coincides with the topological notion if and only if κ is countable or inaccessible, as was shown in [BHZ, §4]. Even though the ideals are distinct if κ is uncountable but not inaccessible, it is an open question whether the cardinal invariants given by the combinatorial κ -meagre ideal are consistently distinct from those given by the topological κ -meagre ideal:

Question 2.3. *Let the combinatorially κ -meagre ideal $\mathcal{M}_\kappa^{\text{comb}}$ be defined as in [BHZ, Def. 4.4]. Is $\text{non}(\mathcal{M}_\kappa^{\text{comb}}) < \text{non}(\mathcal{M}_\kappa)$ or $\text{cov}(\mathcal{M}_\kappa) < \text{cov}(\mathcal{M}_\kappa^{\text{comb}})$ consistent for κ that are not inaccessible?*

The characterisation of $\text{cof}(\mathcal{N})$ and $\text{add}(\mathcal{N})$ from Theorem 2.1 cannot be generalised to the higher context, not just because we would need a higher analogue of $\mathcal{N}$, but also because the cardinality of $\mathfrak{d}_\kappa^h(\in^*)$ is highly dependent on the choice of $h \in {}^\kappa\kappa$ when κ is inaccessible, as we will discuss in Section 3.2. For successor κ , Theorem 2.1 looks very different:

Theorem 2.4 ([Hyt06], [MS, Thm. 4.6]). *If κ is a regular successor, then $\mathfrak{b}_\kappa(\neq^\infty) = \mathfrak{b}_\kappa^h(\in^*) = \mathfrak{b}_\kappa$ and $\mathfrak{d}_\kappa = \mathfrak{d}_\kappa^h(\in^*) \geq \mathfrak{d}_\kappa(\neq^\infty)$. If we also assume $\kappa^{<\kappa} = \kappa$, then the last inequality becomes an equality.*

Question 2.5 ([MS, §4]). *Is $\mathfrak{d}_\kappa(\neq^\infty) < \mathfrak{d}_\kappa$ consistent for κ a successor cardinal and $\kappa^{<\kappa} > \kappa$?*

It was observed by Truss and Miller that the cardinal characteristics $\text{add}(\mathcal{M})$ and $\text{cof}(\mathcal{M})$ can be expressed in terms of the other cardinals of the Cichoń diagram.

Theorem 2.6 ([Tru77], [Mil81]). *$\text{add}(\mathcal{M}) = \min\{\mathfrak{b}, \text{cov}(\mathcal{M})\}$ and $\text{cof}(\mathcal{M}) = \max\{\mathfrak{d}, \text{non}(\mathcal{M})\}$.*

The same characterisation generalises to the higher Cichoń diagram if we make strong enough assumptions on κ . We note that this is not provable for all uncountable κ . For example, Brendle [Bre22] showed that $\kappa^{<\kappa} < \text{cof}(\mathcal{M}_\kappa)$, and thus $\max\{\mathfrak{d}_\kappa, \text{non}(\mathcal{M}_\kappa)\} < \text{cof}(\mathcal{M}_\kappa)$ is consistent, and holds whenever $\max\{\mathfrak{d}_\kappa, \text{non}(\mathcal{M}_\kappa)\} \leq \kappa^{<\kappa}$.

Theorem 2.7 ([Bre22]). *$\text{add}(\mathcal{M}_\kappa) = \min\{\mathfrak{b}_\kappa, \text{cov}(\mathcal{M}_\kappa)\}$ and $\text{cof}(\mathcal{M}_\kappa) \geq \max\{\mathfrak{d}_\kappa, \text{non}(\mathcal{M}_\kappa)\}$. Under assumption of $\kappa^{<\kappa} = \kappa$ the inequality becomes an equality.*

As for the other eight cardinal characteristics of the classical Cichoń diagram, they appear to be as independent as is possible in the following sense: any assignment of the cardinalities $\aleph_1$ and $\aleph_2$ to the cardinals of the classical Cichoń diagram compatible with both Theorem 2.6 and the arrows in the diagram, is consistent (see [BJ95, Sections 7.5 and 7.6] for each of these cases). This implies that the classical Cichoń diagram is complete, as no arrows are missing from the diagram. In fact, Goldstern, Kellner, Mejía, and Shelah [GKMS21; GKS19] showed that all eight cardinal characteristics can have mutually different cardinalities within the same model.